

Baibhav Singh

Kathmandu, NP | +977.984007.7200 | singh.baibhav321@gmail.com | [LinkedIn](#) | [Portfolio](#) | [Github](#)

OBJECTIVE

Senior QA Automation Engineer with 4+ years of industry experience in test automation, AI-integrated tooling, and CI/CD pipelines, seeking admission to a graduate program to deepen expertise in software engineering, intelligent systems, and distributed computing, and to apply research skills toward building reliable, large-scale software solutions.

EDUCATION

[Vellore Institute of Technology](#), Tamil Nadu, IND

June 2017 – July 2021

- Bachelor of Technology in Computer Science and Engineering | CGPA: 7.91 / 10.

APS Certificate (Akademische Prüfstelle)

2026

- Academic credentials successfully evaluated by the German Academic Evaluation Centre ([APS](#)) | Prüfnummer: 03875/26.

PROJECTS

[Intelligent Healthcare Monitoring System with Pulse Oximetry Analysis](#)

Final Year Project | 2021

Vellore Institute of Technology | Machine Learning, Arduino, IoT, Python

- Built an IoT health monitoring system using Arduino Uno for real-time pulse oximetry data acquisition, applying supervised ML models to classify SpO₂ and heart rate readings across multiple risk levels.
- Implemented an end-to-end pipeline from embedded sensor data collection through ML inference, demonstrating cross-disciplinary integration of hardware, software, and intelligent systems.

[Smart Stick for Visually Impaired](#)

Academic Project | 2020

Vellore Institute of Technology | Arduino, C++, IoT, GPS, GSM (SIM808)

- Developed a sensor-based assistive navigation device using ultrasonic and water sensors to detect obstacles, puddles, and hazards in real time, enabling independent mobility for visually impaired users.
- Integrated a SIM808 GPS/GSM module to transmit the user's live location to a caregiver via Google Maps SMS at regular intervals, with a physical SOS button triggering an emergency alert for immediate assistance.

SKILLS

- **Framework:** Katalon, Selenium, Cypress, Playwright, SoapUI, REST Assured, TestNG, JUnit, NodeJS, CUDA.
- **Tools:** Katalon Studio, Postman, Git, Jenkins, Jira, Zephyr, Bitbucket, Confluence, VS Code.
- **Language:** Groovy, Java, SQL, JavaScript, Python, C++, C.
- **QA:** Software Testing, Requirements Analysis, Test Planning, Bug Life Cycle, Root Cause Analysis, CI/CD, STLC, SDLC.
- **Relevant Coursework:** Data Structures and Algorithms, Discrete Mathematics and Graph Theory, Calculus, Linear Algebra, Statistics, Image Processing, High Performance Computing, Database Management Systems, Software Engineering, Network and Communication, Internet of Things, Operating Systems, Microprocessor and Interfacing, Java Programming, Machine Learning.
- **Soft Skills:** Leadership, Critical Thinking, Research, Ownership, Analytical, Collaboration, and Teamwork.
- **Spoken:** English (Professional), Nepali (Native), Hindi (Fluent).

EXPERIENCE

[Propertystack Software Pty Ltd](#), Brisbane, Australia

November 2025 – Present

Senior QA Engineer

- **Built a Playwright automation framework from scratch** incorporating AI Agents for test planning, script generation, and self-healing; delivered a 5-step locator strategy and 100+ reusable page functions with tests executed in parallel.
- **Integrated the Kovenia payment system** across 2 countries, validating end-to-end payment flows through both UI and API testing with zero security vulnerabilities found at release.

- **Executed 500+ test cases per feature cycle**, catching 150+ bugs per feature and maintaining a high defect-detection rate ahead of each release.
- **Validated a ChatGPT-powered in-app chatbot** across 3 sentiment levels, verifying response accuracy against live application data to ensure contextually relevant outputs.
- **Ensured integrity of complex business workflows** spanning payments, emails, and 10+ rule combinations, validating each against user requirements.
- **Verified trust accounting accuracy for reservation disbursements and reversals** across 100+ scenarios, identifying 100+ bugs to ensure financial correctness and compliance.

[Cedar Gate Technologies](#), Lalitpur, Nepal

June 2023 – December 2025

Software QA Automation Engineer / Payment Technology Automation QA

- **Scaled test coverage from 29% to 82%** for a healthcare SaaS payment platform, independently automating 100+ test cases and increasing script reusability by 25% through template libraries.
- **Reduced test script maintenance by 60%** via Excel-based data-binding and evaluated 10 frameworks, implementing the optimal stack to cut development time by 20% via CI/CD.
- **Mentored 5 junior engineers**, collectively automating 75% of 15,000 test cases within one year and authored onboarding guides for smooth knowledge transfer.

[Cedar Gate Technologies](#), Lalitpur, Nepal

November 2021 – June 2023

Software QA Engineer / Payment Technology QA

- **Identified and remediated 600+ SQL injection vulnerabilities** using JIRA-integrated workflows, significantly strengthening application security posture.
- Executed regression, smoke, and sanity testing across 12 consecutive production releases, ensuring quality across every delivery cycle.
- Authored 10+ QA guidelines and process manuals adopted by a 20-member team, establishing org-wide standards for testing and defect reporting.

CERTIFICATIONS

- [IELTS Academic](#) — Overall Band 7.0 | English language proficiency for graduate study.
- [Supervised Machine Learning](#), Coursera — Supervised ML models with 80% accuracy gain and 45% classification efficiency boost.
- [Divide and Conquer, Sorting and Searching, and Randomized Algorithms](#) — Core algorithm design and analysis for software engineering.
- [Fundamentals of Parallelism on Intel Architecture](#) — Multi-core and parallel programming on Intel architecture for performance-critical code.
- [QA Automation](#) — Keyword-driven Selenium framework with TestNG, data-driven testing, and Jenkins CI/CD.

VOLUNTEER WORK / ACTIVITIES

[Cedar Gate Technologies](#), Professional Communication Club

January 2023 – December 2023

Vice-President

- Coordinated 25 Toastmasters sessions for ~600 Cedar Gate Nepal employees (200 active participants), alongside 2 panel discussions and 24 knowledge-sharing articles.

RESEARCH INTERESTS

Intelligent Systems · Robotics · Parallel & High-Performance Computing (CUDA) · AI-Driven Software Engineering